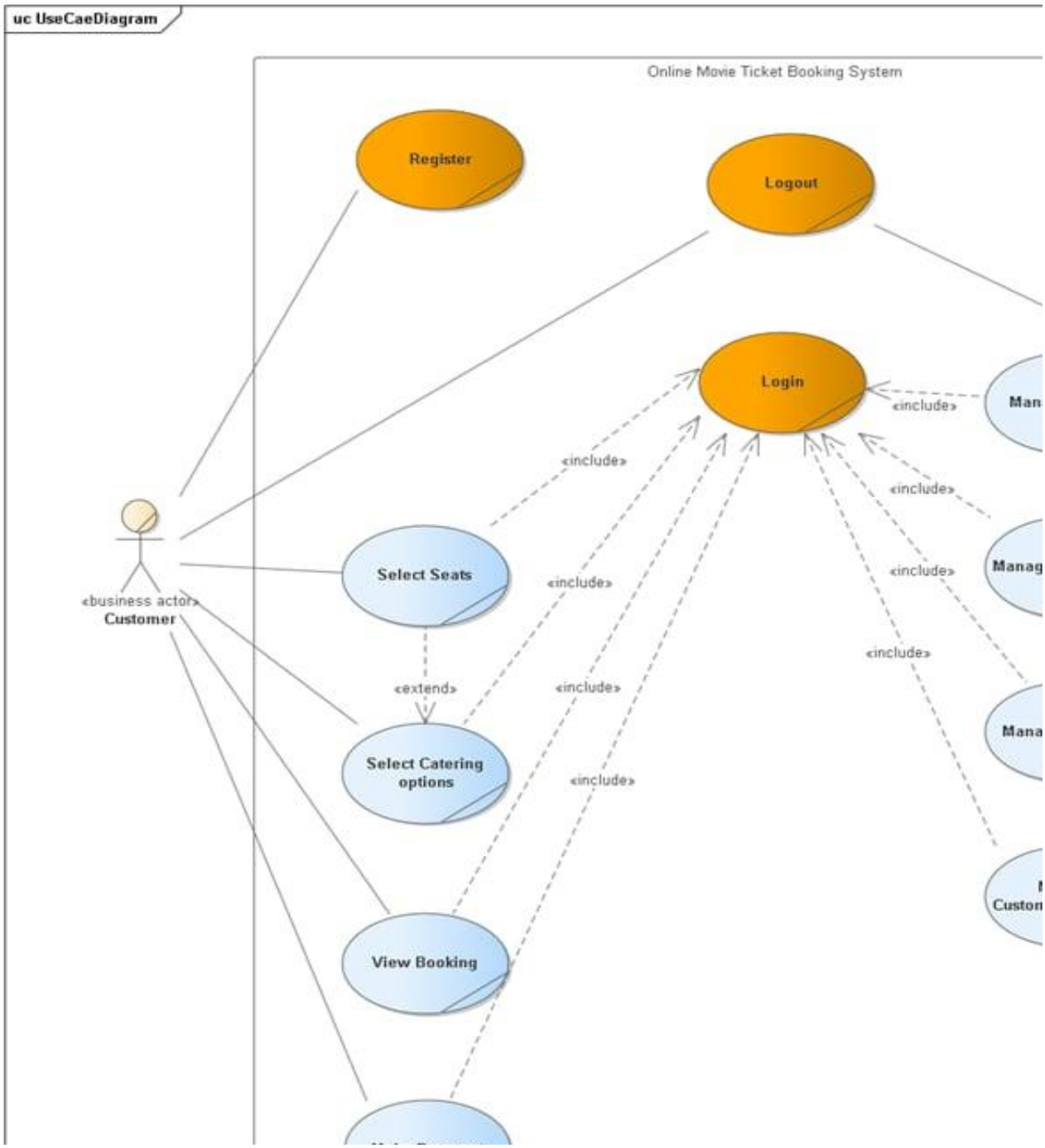
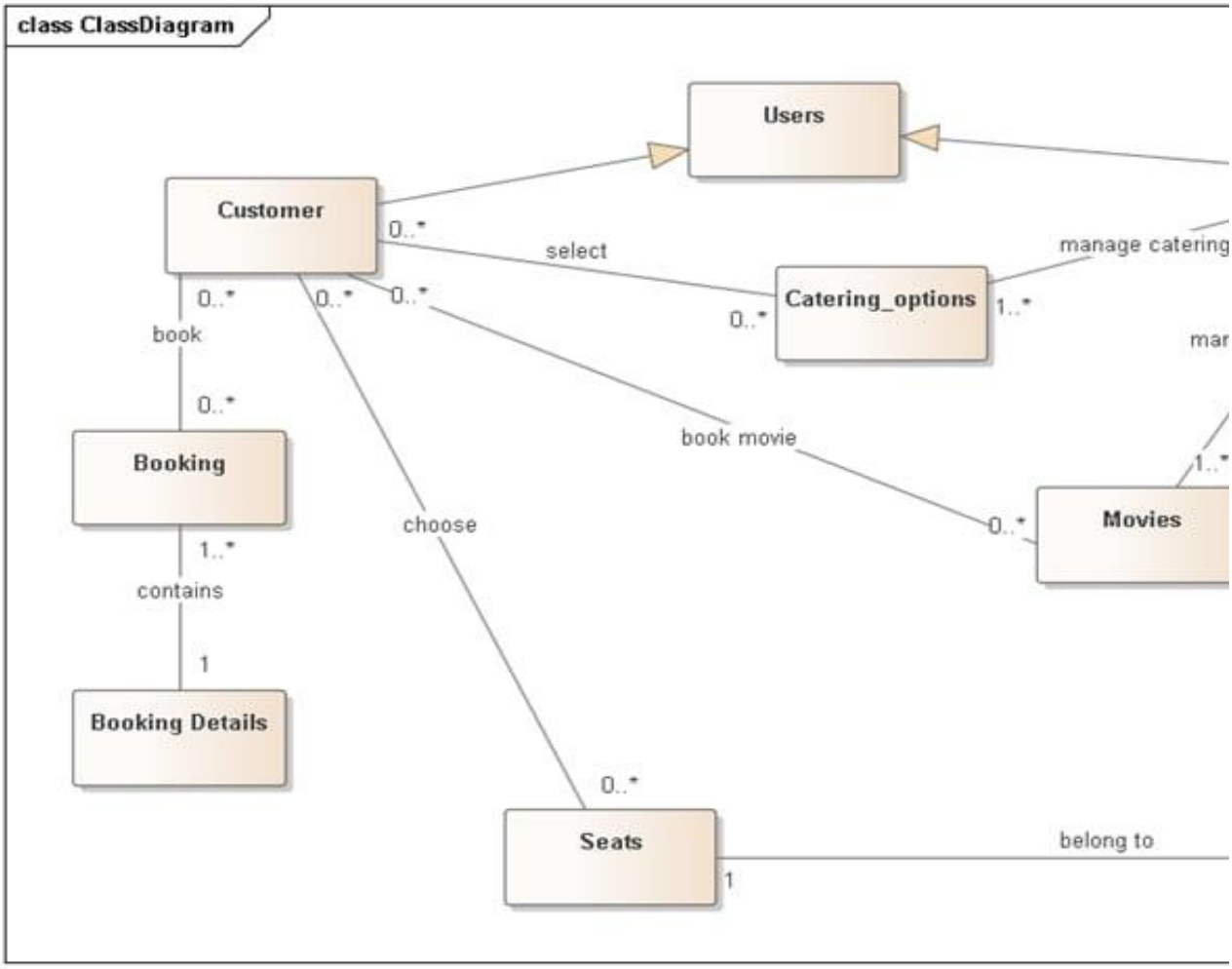
Student name
Trần Ngọc Ánh Dương
Ao Thành Nhân
Lưu Dương Bá Bảo

Online Movie Ticket Booking System - Do

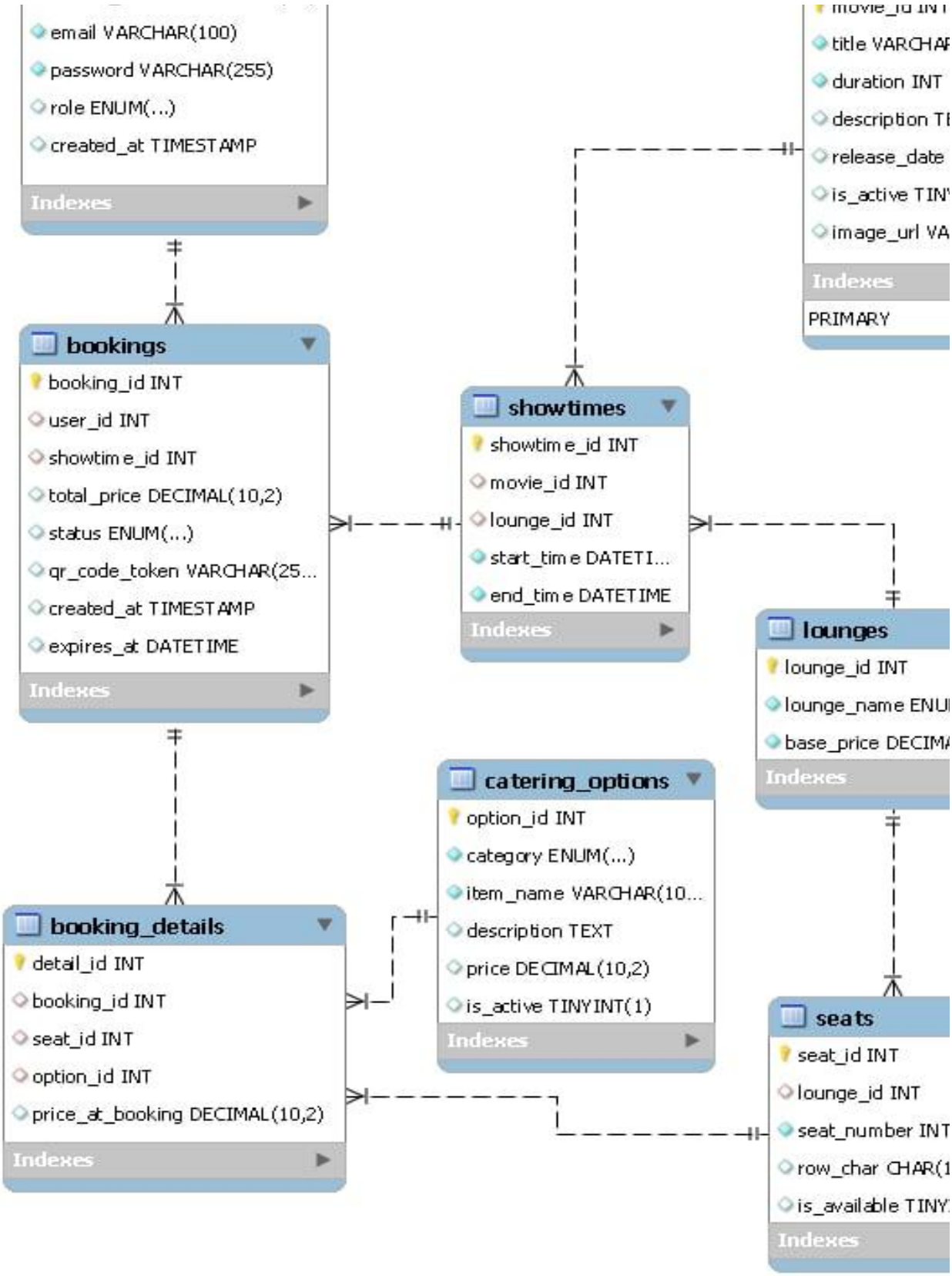
Usecase



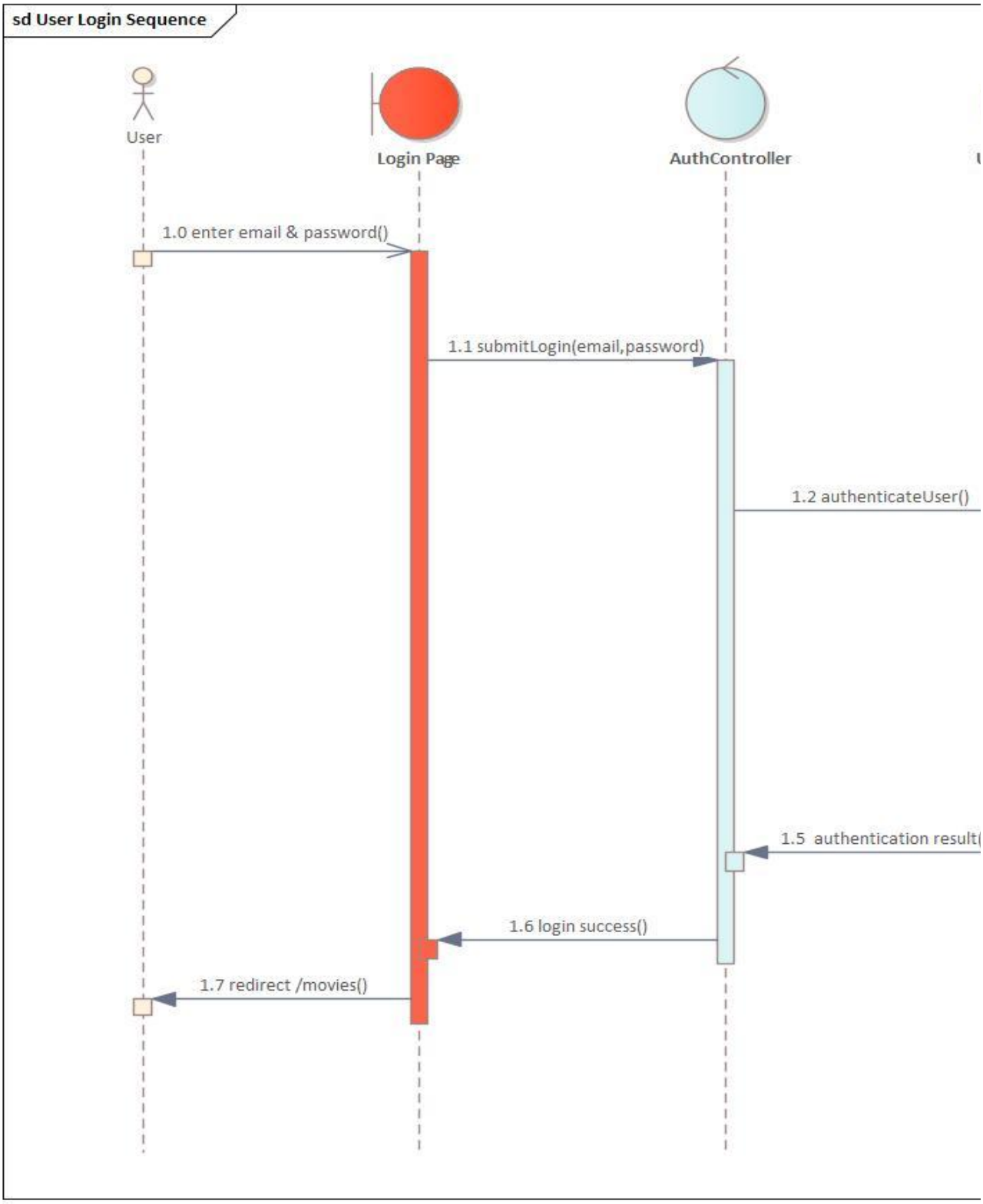
ERD



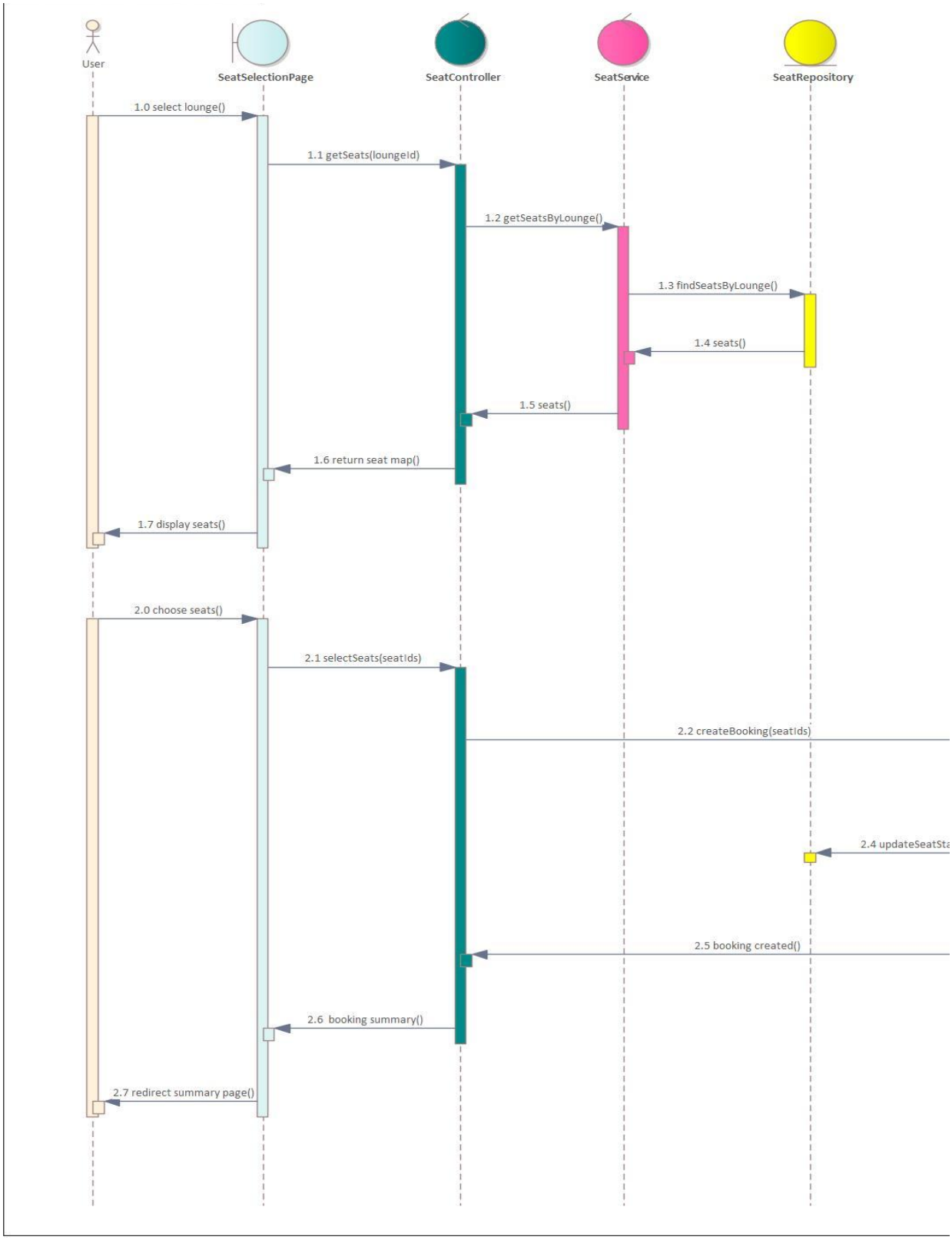
Database Design

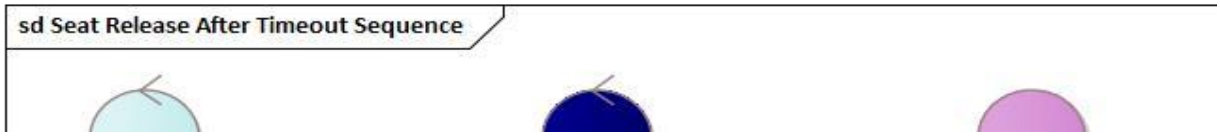

email VARCHAR(100)
password VARCHAR(255)
role ENUM(...)
created_at TIMESTAMP
Indexes
title VARCHAR
duration INT
description TE
release_date
is_active TIN
image_url VA
Indexes
PRIMARY
bookings
booking_id INT
user_id INT
showtime_id INT
total_price DECIMAL(10,2)
status ENUM(...)
qr_code_token VARCHAR(25...
created_at TIMESTAMP
expires_at DATETIME
Indexes
showtimes
showtime_id INT
movie_id INT
lounge_id INT
start_time DATETI...
end_time DATETIME
Indexes
lounges
lounge_id INT
lounge_name ENU
base_price DECIMA
Indexes
catering_options
option_id INT
category ENUM(...)
item_name VARCHAR(10...
description TEXT
price DECIMAL(10,2)
is_active TINYINT(1)
Indexes
booking_details
detail_id INT
booking_id INT
seat_id INT
option_id INT
price_at_booking DECIMAL(10,2)
Indexes
seats
seat_id INT
lounge_id INT
seat_number INT
row_char CHAR(1
is_available TINY
Indexes

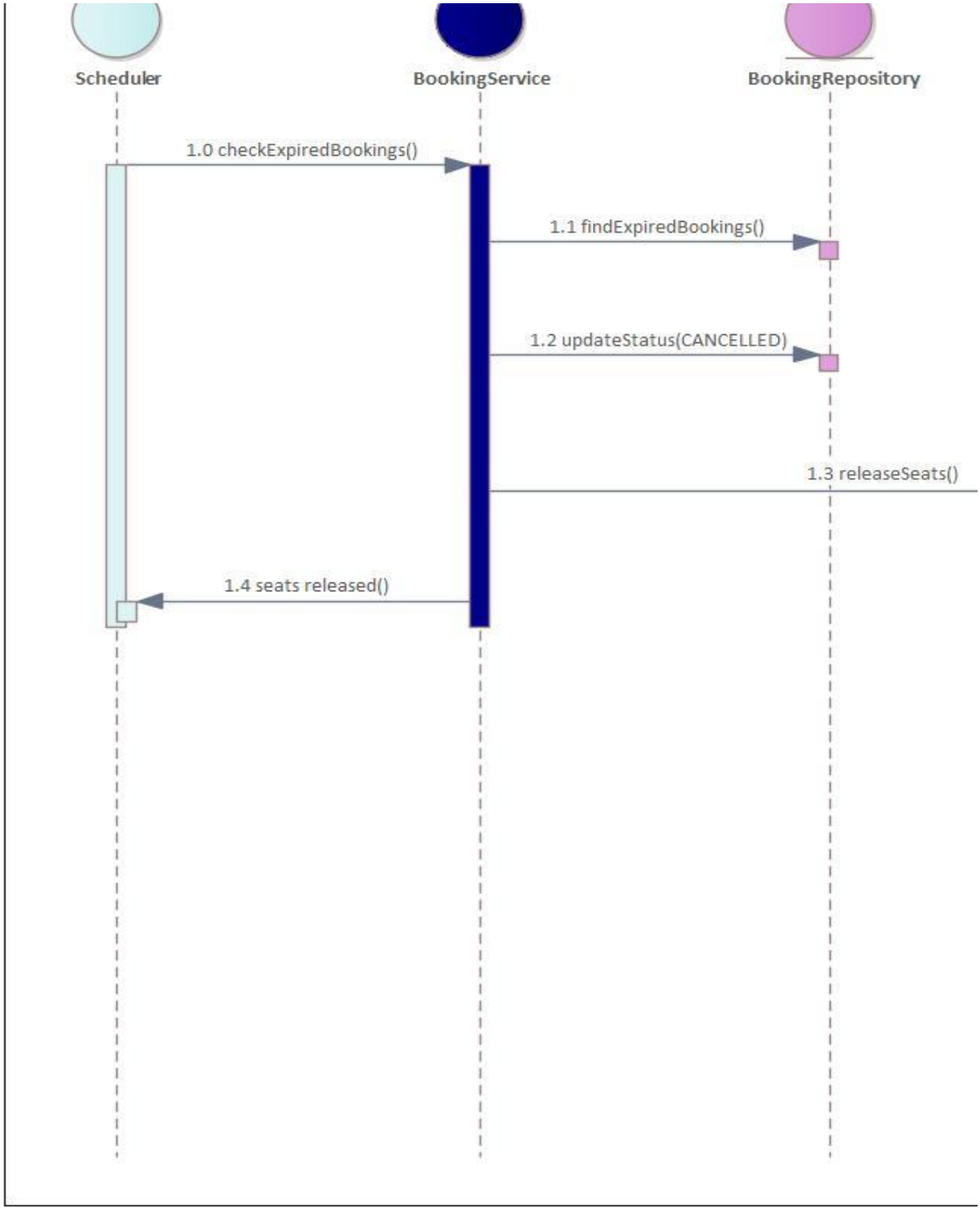
Sequence

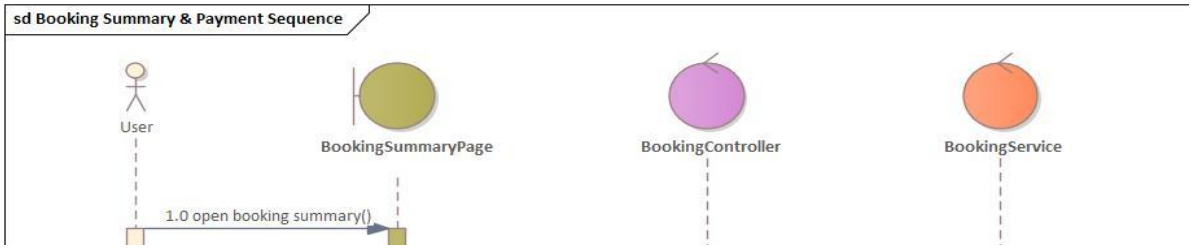

sd User Login Sequence
User
Login Page
AuthController
1.0 enter email & password()
1.1 submitLogin(email,password)
1.2 authenticateUser()
1.5 authentication result(
1.6 login success()
1.7 redirect /movies()

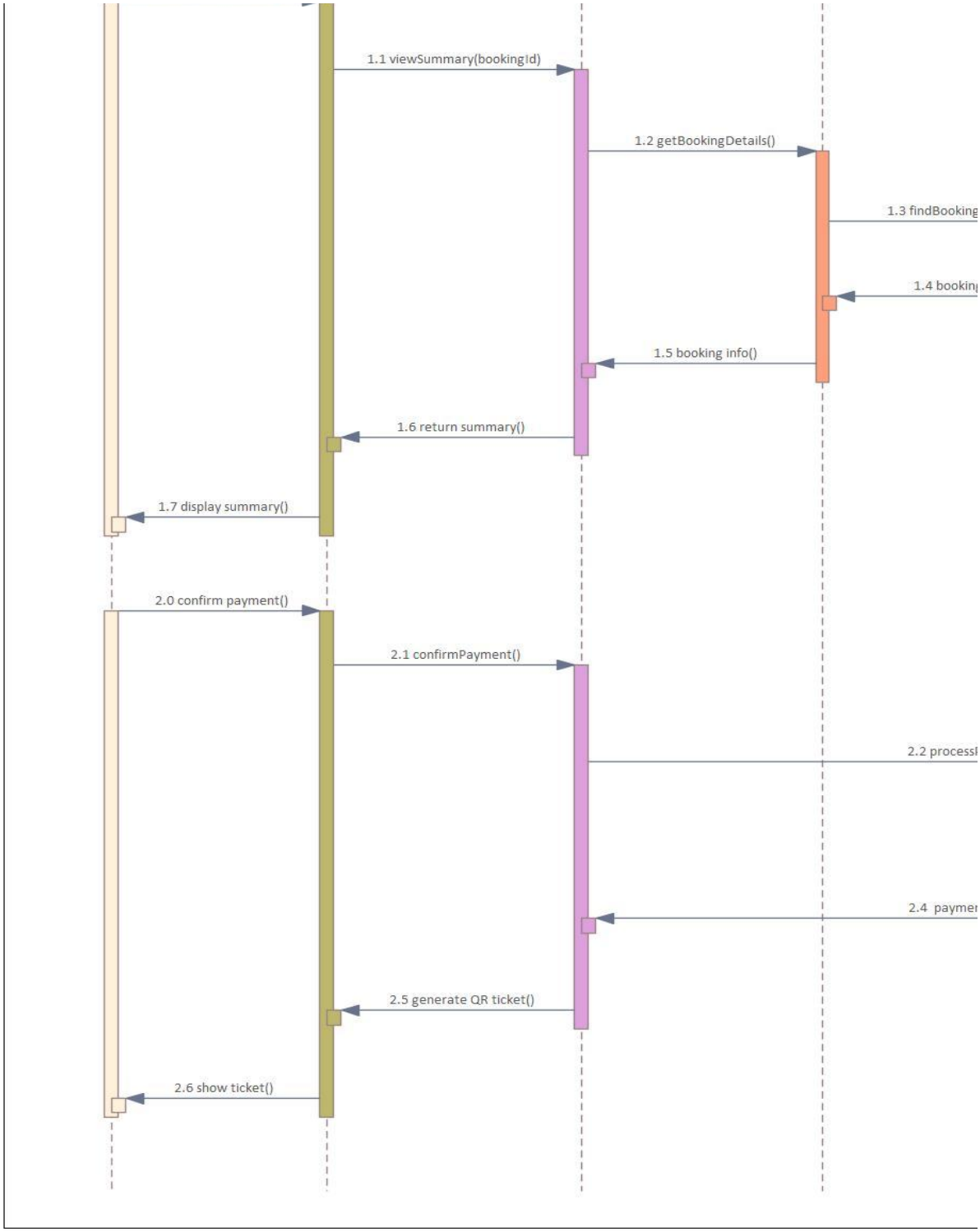
sd Seat Selection & Booking Creation Sequence

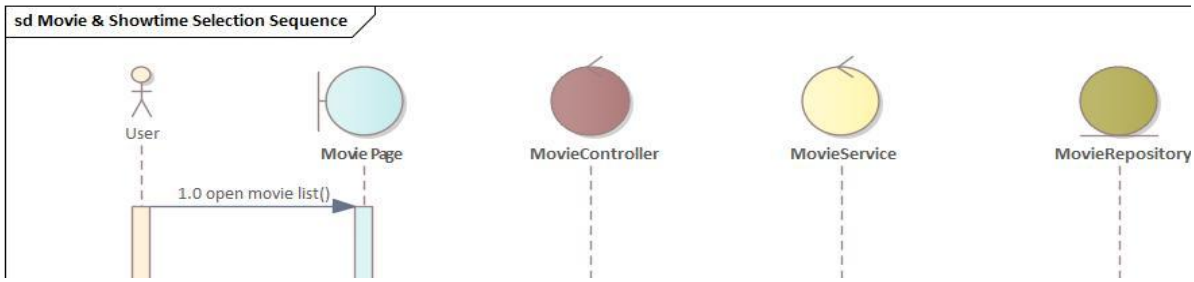

User
SeatSelectionPage
SeatController
SeatService
SeatRepository
1.0 select lounge()
1.1 getSeats(loungeId)
1.2 getSeatsByLounge()
1.3 findSeatsByLounge()
1.4 seats()
1.5 seats()
1.6 return seat map()
1.7 display seats()
2.0 choose seats()
2.1 selectSeats(seatIds)
2.2 createBooking(seatIds)
2.4 updateSeatSta
2.5 booking created()
2.6 booking summary()
2.7 redirect summary page()

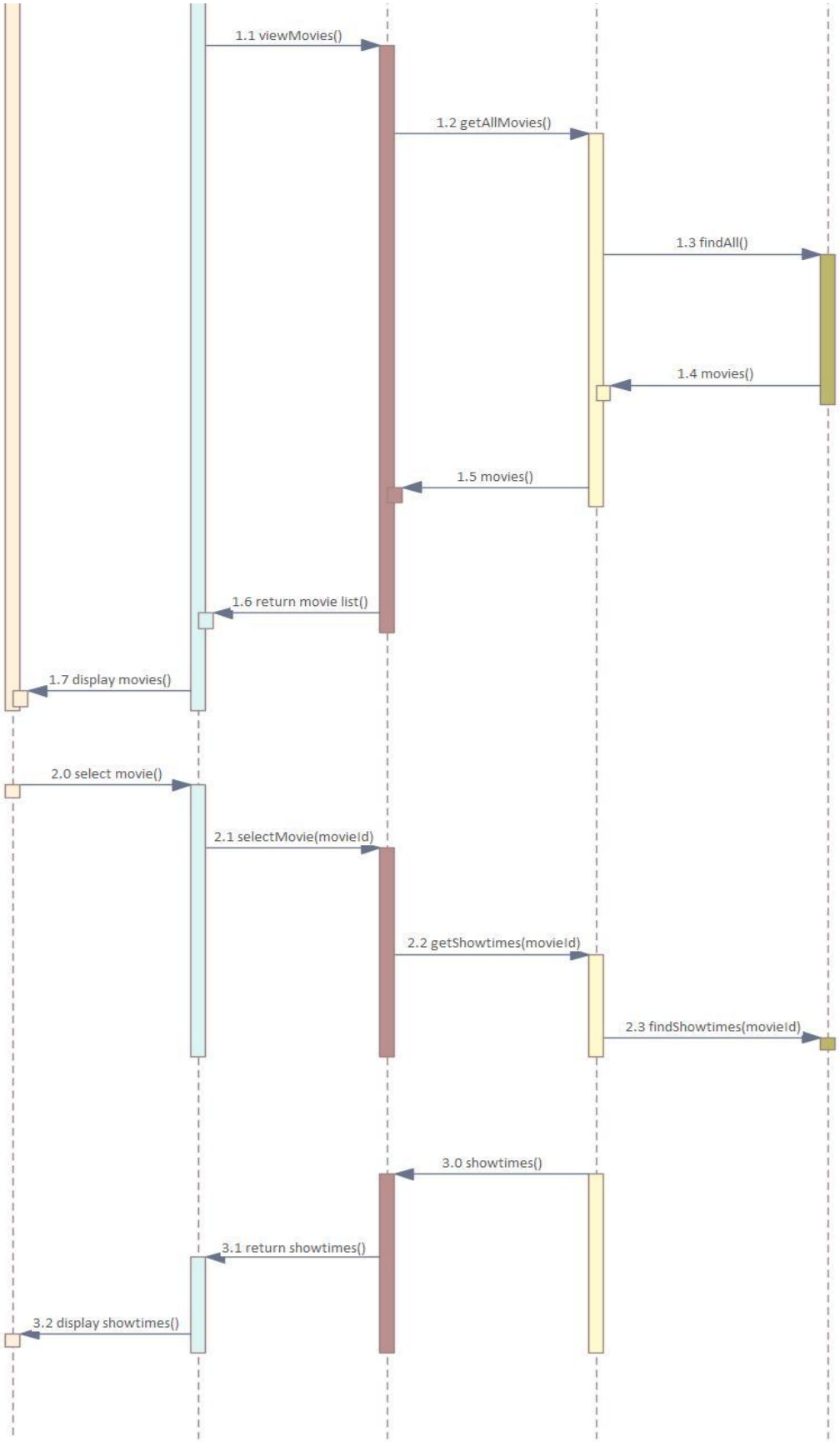

sd Seat Release After Timeout Sequence


Scheduler
BookingService
BookingRepository
1.0 checkExpiredBookings()
1.1 findExpiredBookings()
1.2 updateStatus(CANCELLED)
1.3 releaseSeats()
1.4 seats released()


sd Booking Summary & Payment Sequence
User
BookingSummaryPage
BookingController
BookingService
1.0 open booking summary()

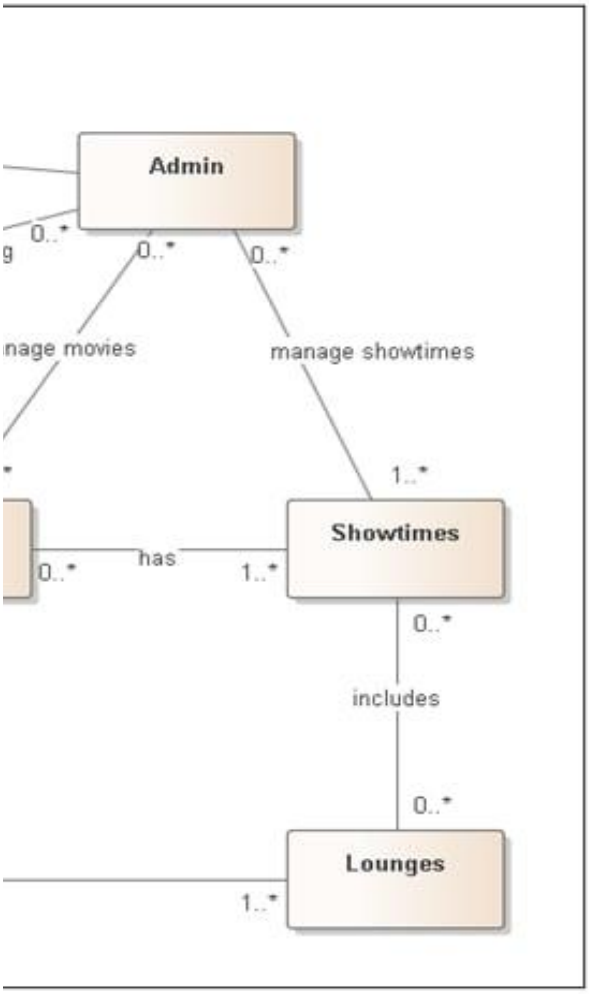

1.1 viewSummary(bookingId)
1.2 getBookingDetails()
1.3 findBooking
1.4 booking
1.5 booking info()
1.6 return summary()
1.7 display summary()
2.0 confirm payment()
2.1 confirmPayment()
2.2 processI
2.4 paymer
2.5 generate QR ticket()
2.6 show ticket()


sd Movie & Showtime Selection Sequence
User
Movie Page
MovieController
MovieService
MovieRepository
1.0 open movie list()


1.1 viewMovies()
1.2 getAllMovies()
1.3 findAll()
1.4 movies()
1.5 movies()
1.6 return movie list()
1.7 display movies()
2.0 select movie()
2.1 selectMovie(movieId)
2.2 getShowtimes(movieId)
2.3 findShowtimes(movieId)
3.0 showtimes()
3.1 return showtimes()
3.2 display showtimes()

Student ID	Class
2280600534	22DTHQB2
2280600194	22DTHQB2
2280602150	22DTHQB2

ɔcumentation Report


Admin
0..*
0..*
0..*
nage movies
manage showtimes
1..*
Showtimes
0..*
has
1..*
0..*
includes
0..*
Lounges
1..*

R(255)

EXT

DATE

YINT(1)

ARCHAR(500)

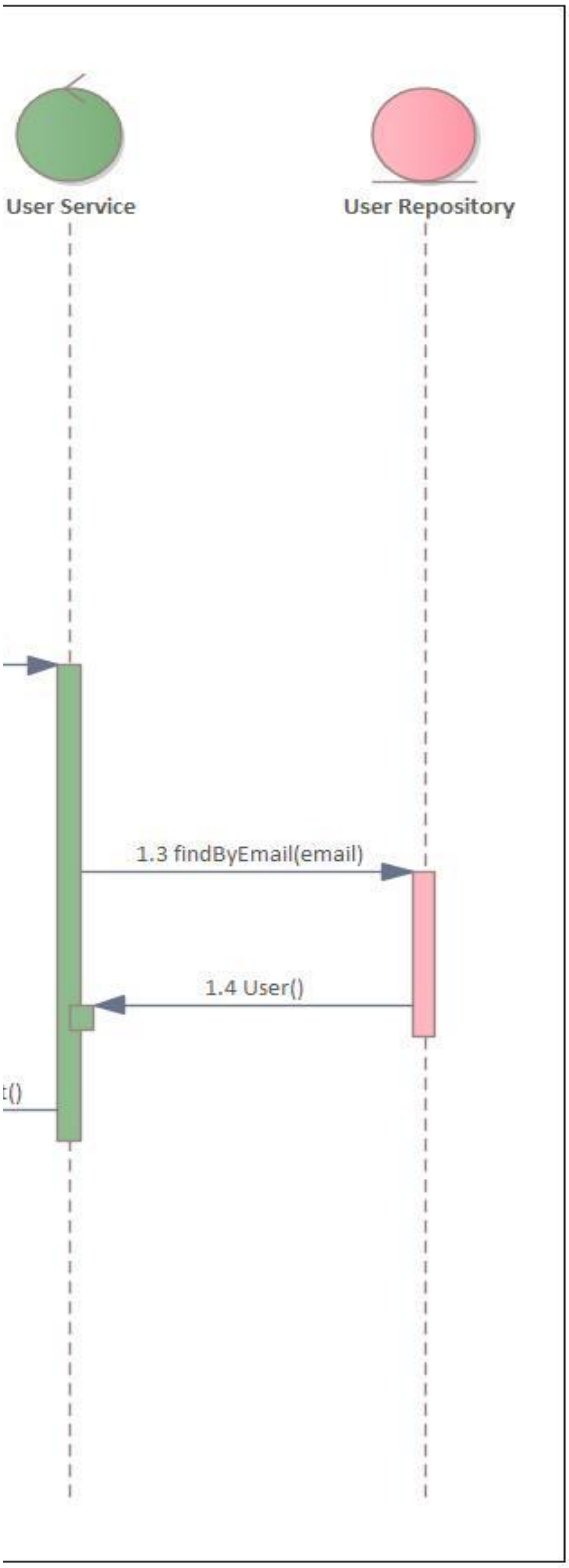
JM(...)

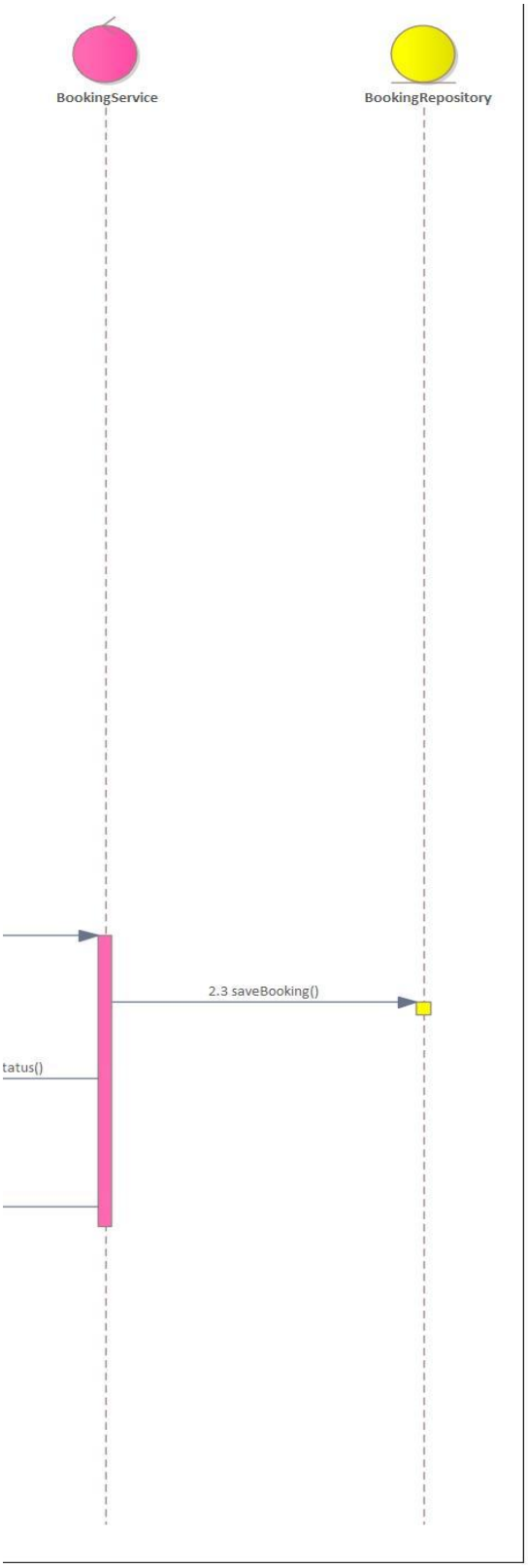
AL(10,2)

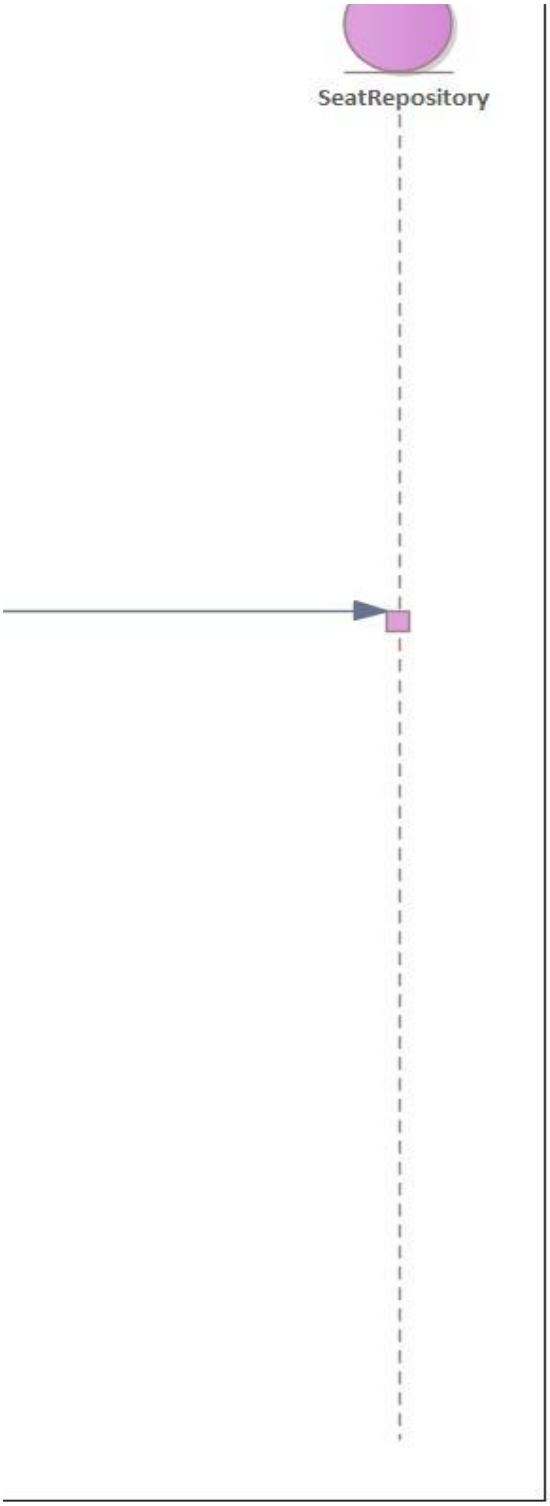
T

1)

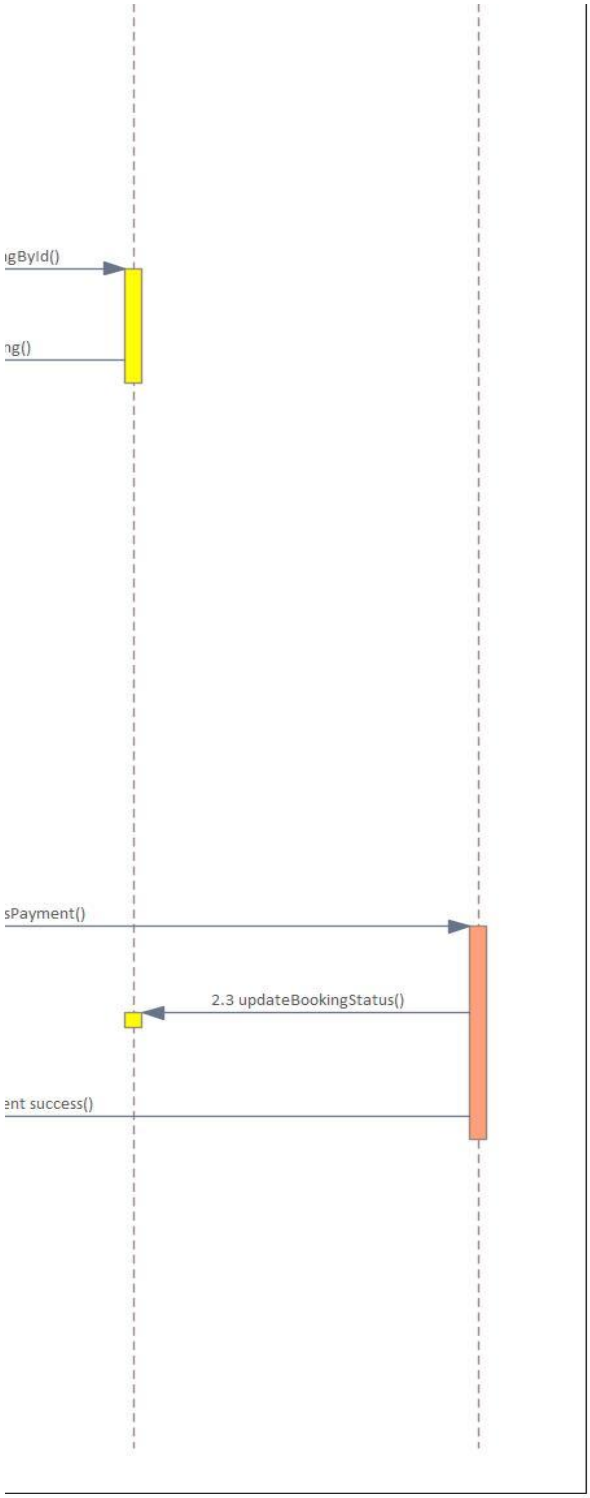
/INT(...


User Service
User Repository
1.3 findByEmail(email)
1.4 User()


BookingService
BookingRepository
2.3 saveBooking()
tatus()


SeatRepository

BookingRepository
PaymentService



y

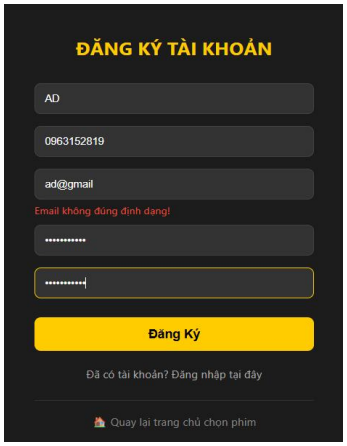
Login & Authentication

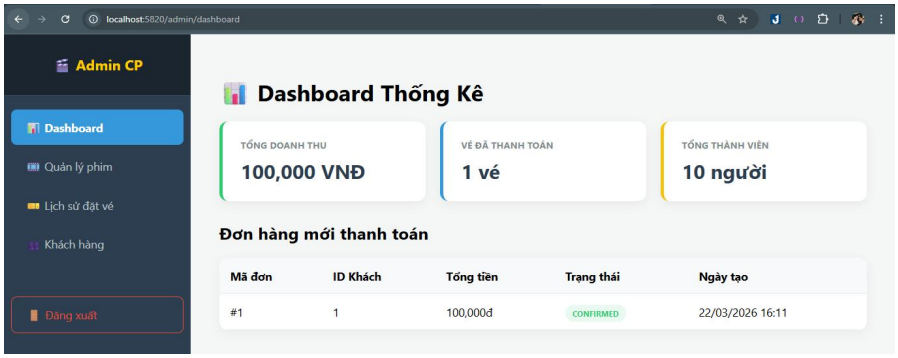
STT	Method	Test Scenario
1	passwordEncoder	Test BCrypt password encryption
2	isValidEmail	Check email format validation
3	loadUserByUsername	Check role authorization after login

Input	Expected Result	Status
"Anhduong123"	Returns a hashed string (not "Anhduong123")	Pass
"nhan@gmail" (missing .com	Returns False	Pass
Email: admin1@gmail.com	Returns a User with ROLE_ADMIN	Pass

Image

email	password
admin@gmail.com	123456
thinh@gmail.com	$2a$10$a72mcEWMD.QDQIYDeWWR0Ogf/1bgDrmlVQFdc/AW8yQUwfwr/0...
nhan@gmail.com	$2a$10$s1XNx/917x9manLQ6xZFkOuRt3kFUdyDRf/lyEtY1zBGARgVQp6bK
admin_vip@gmail.com	$2a$10$irJS1x3k2WWGqj8bCPnolO0RNQp2yTJuyIflEyQ88IPQeezxejlgC
admin1@gmail.com	$2a$10$7x/DTT89LZiEPg6FKTQBxuMEr0UvRV1IZ3/NOg0YvXDtt1Z.V932C
duy@gmail	$2a$10$GvP5vSNIuuVeGrrrnCsJU.KTi3Q6NLlM2NPSKziAoTawH4Ts2la52
ad@gmail.com	$2a$10$oaigl59SzRkEzs72FoZPC.oIAPA0BbLtyzrDXxX/oW1w4adyliRcy





Select Showtime & Lounge

STT	Method	Test Scenario
1	selectShowtime()	Filter the list of showtimes based on the selected movie
2	selectLounge()	User selects a lounge for the chosen showtime
3	autoSeat()	The system automatically suggests the best available seat

Input

MovieID: 1

ShowtimeID: 1

LoungeID: 1

Expected Result

The system displays all showtimes belonging to MovieID = 1

The system stores currentShowtimeId in Session and displays the lounge selection page

The system returns the best seat (priority for rows E, F, G and center seats)

Status

Pass

Pass

Pass

Seat Selection

STT	Method	Test Scenario
1	getSeatsByLounge()	Retrieve the list of seats for a lounge
2	showSeatMap()	Display the seat map for user selection
3	autoAssignSeat()	System automatically selects an available seat
4	createMultipleBooking()	User selects multiple seats and creates a booking
5	saveSeatStatus()	Update seat status after booking

Input

LoungeID: 1

LoungeID: 2

LoungeID: 1

SeatIDs: [G65, H66, H67]
SeatID: 15

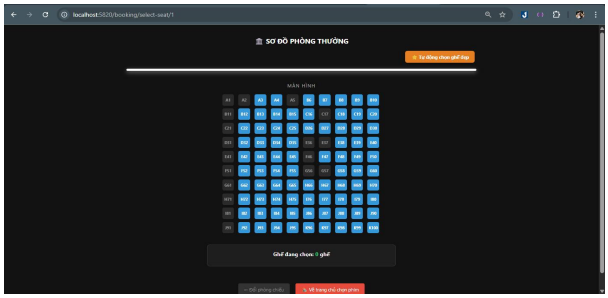
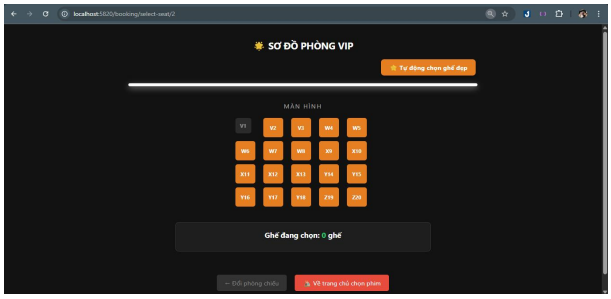
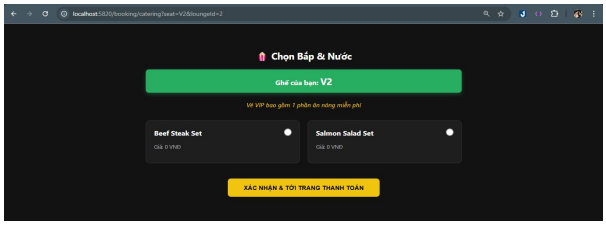
Expected Result

The system returns all seats belonging to Lounge 1

The seat-selection page displays the seats of the VIP lounge

The system returns the first available seat or a center seat

The system creates a new Booking and locks the selected seats
The database updates is_available = 0

Status	Image
Pass	
Pass	
Pass	
Pass	
Pass	

Spring Security Authentication

STT	Method	Test Scenario
1	Access Booking Page	Access /booking/select-seat without login
2	User Login	Login with a valid user account
3	Admin Login	Login with an admin account
4	Unauthorized Access	User attempts to access /admin

5	Logout	User logs out

Input

No login

Email = nhan@gmail.com , pass word = Nhan123456

Admin email = "admin1@gmail.com", password="Admin1"

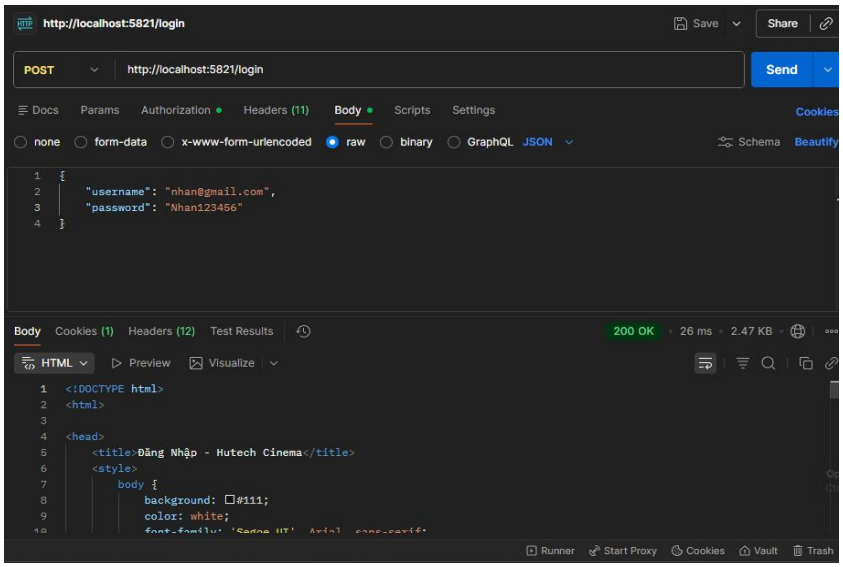
User account

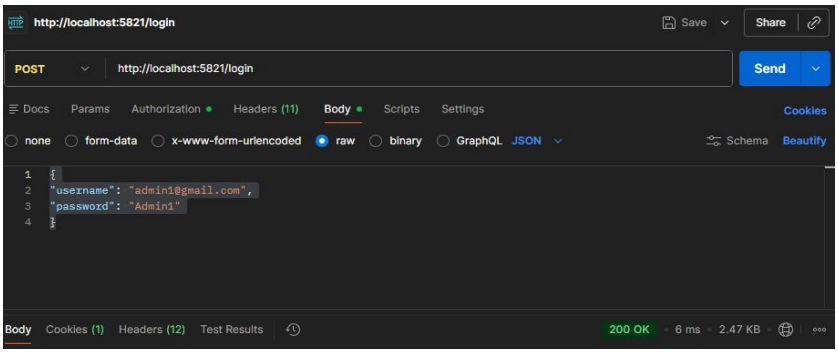
Click Logout

Expected Result	Status
Redirect to /login	Pass
Redirect to /movies	Pass
Redirect to /admin/dashboard	Pass
Access Denied	Pass

Redirect to /movies	Pass

Image


http://localhost:5821/login
Save
Share
POST
http://localhost:5821/login
Send
Docs
Params
Authorization
Headers (11)
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
{
"username": "nhan@gmail.com",
"password": "Nhan123456"
}
Body
Cookies (1)
Headers (12)
Test Results
200 OK
26 ms
2.47 KB
HTML
Preview
Visualize
<!DOCTYPE html>
<html>
<head>
<title>Đăng Nhập - Hutech Cinema</title>
<style>
body {
background: #111;
color: white;
Runner
Start Proxy
Cookies
Vault
Trash


http://localhost:5821/login
Save
Share
POST
http://localhost:5821/login
Send
Docs
Params
Authorization
Headers (11)
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
{
"username": "admin1@gmail.com",
"password": "Admin1"
}
Body
Cookies (1)
Headers (12)
Test Results
200 OK
6 ms
2.47 KB

http://localhost:5821/logout

Save | Share

POST http://localhost:5821/logout Send

Docs | Params | Authorization | Headers (11) | Body | Scripts | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

```
1 {
2 "username": "nhan@gmail.com",
3 "password": "Nhan123456"
4 }
```

Body | Cookies (1) | Headers (12) | Test Results | 200 OK | 13 ms | 3.79 KB

XML | Preview | Visualize

```
1 <!DOCTYPE html>
2 <html>
3     <head>
4         <meta charset="UTF-8">
5             <title>Online Movie Ticket</title>
6             <style>
7         body { margin:0; font-family: 'Segoe UI', Arial; background:#111; color:white; }
8         header { display:flex; justify-content:space-between; align-items:center; padding:15px 40px;
              background:#000; position: sticky; top:0; z-index: 100; }
```

Runner | Start Proxy | Cookies | Vault | Trash

Booking Summary & Payment

STT	Method	Test Scenario
1	summary()	Generate the booking summary page
2	calculateTotalPrice()	Calculate the total price for tickets and food
3	sepayWebhook()	Receive payment webhook from the bank
4	checkStatus()	Check payment status

5	cancelAndReleaseSeat()	Cancel booking and release seats

Input

Seat: E45, LoungeID: 1, Food: 3

Ticket: 75000, Food: 0

Content: "HUTECH 32", Amount: 75000

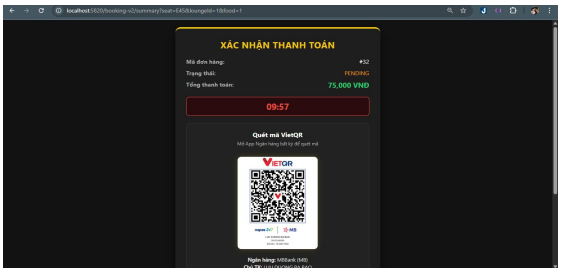
BookingID: 105

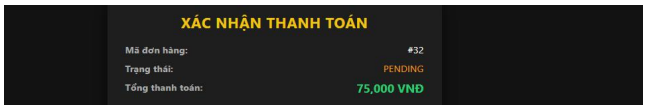
BookingID: 100

Expected Result	Status
The system displays the Booking Summary with a 10-minute seat holding timer	Pass
Total price = 75000	Pass
Booking 32 status is updated to CONFIRMED	Pass
The system returns status CONFIRMED	Pass

Status becomes CANCELLED and seats are released	Pass

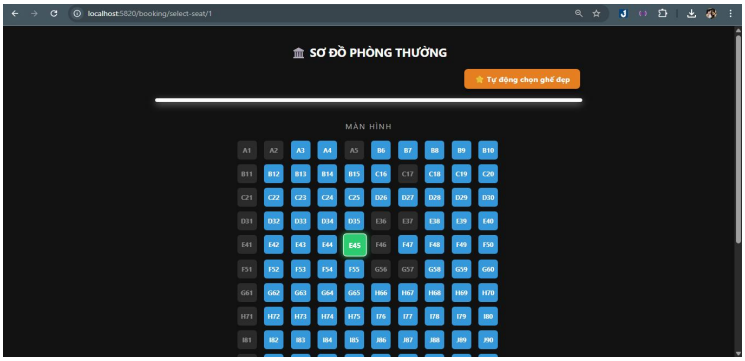
Image


XÁC NHẬN THANH TOÁN
Mã đơn hàng:
#32
Trạng thái:
PENDING
Tổng thanh toán:
75,000 VNĐ
09:57
Quét mã VietQR
VIETQR


XÁC NHẬN THANH TOÁN
Mã đơn hàng:
#32
Trạng thái:
PENDING
Tổng thanh toán:
75,000 VNĐ

Đến
LUU DUONG BA BAO
MB Bank - NH TMCP Quan Doi
0845546888
Số tiền
75.000
VND
Nội dung
HUTECH 32
Tiếp tục

booking_id	user_id	showtime_id	total_price	status	qr_code_token	created_at	expires_at
25	1	1	75000	CANCELLED	NULL	2026-03-26 23:38:23	2026-03-26 23:48:24
26	1	1	75000	CANCELLED	NULL	2026-03-26 23:38:32	2026-03-26 23:48:33
27	1	1	75000	CANCELLED	NULL	2026-03-26 23:51:48	2026-03-27 00:01:49
28	1	1	75000	CANCELLED	NULL	2026-03-26 23:51:55	2026-03-27 00:01:55
29	1	1	75000	CANCELLED	NULL	2026-03-26 23:54:00	2026-03-27 00:04:01
30	1	1	75000	CANCELLED	NULL	2026-03-27 13:06:40	2026-03-27 13:16:40
31	1	1	75000	CANCELLED	NULL	2026-03-27 14:54:31	2026-03-27 15:04:31
32	1	1	75000	CANCELLED	NULL	2026-03-27 15:04:07	2026-03-27 15:14:07



Movie Booking Flow

No.	Flow Step	Test Scenario	Input
1	Open Movie List	User accesses the movie list page	URL: /movies
2	Select Movie	User selects a movie	MovieID: 1
3	Select Showtime	User selects a showtime	ShowtimeID: 1
4	Select Lounge	User chooses Standard or VIP lounge	LoungeID: 1
5	Seat Selection	User selects seats from the seat map	SeatIDs: [E45]
6	Booking Summary	System generates booking summary	BookingID: 33

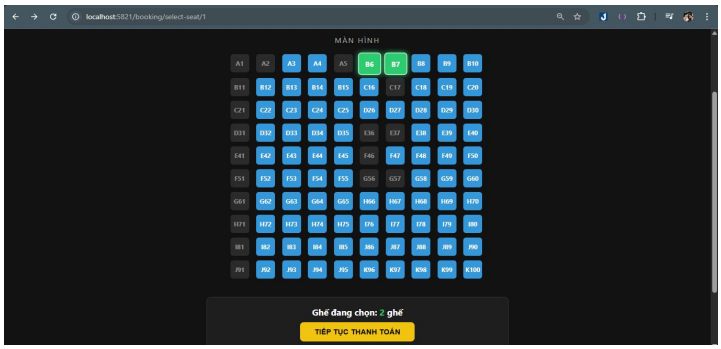
Expected Result	Status
The system displays a list of available movies	Pass
The system loads showtimes of the selected movie	Pass
The system redirects to the lounge selection page	Pass
The system loads the seat selection page	Pass
The system temporarily locks the selected seats	Pass
Summary page displays ticket information and price	Pass

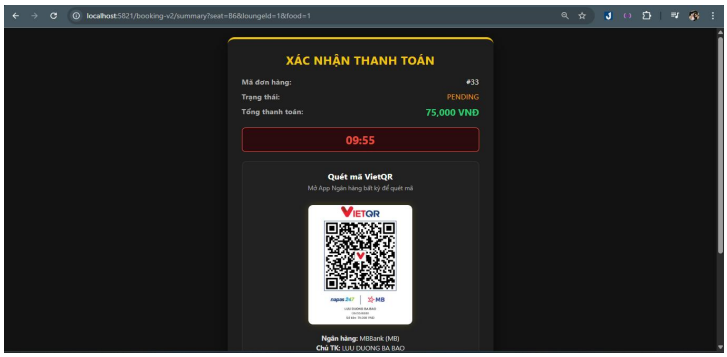
Seat Reservation Flow

No.	Flow Step	Test Scenario	Input
1	Load Seat Map	System loads seat layout for the lounge	LoungeID: 1
2	Select Seats	User selects multiple seats	SeatIDs: [B6, B7]
3	Seat Validation	System checks seat availability	SeatIDs: [B6, B7]
4	Temporary Lock	System holds the seats for booking	SeatIDs: [B6, B7]

Expected Result	Status
The seat map is displayed with available and reserved seats	Pass
Seats are marked as selected	Pass
System confirms seats are available	Pass
Seats are temporarily locked for 10 minutes	Pass

Image


localhost:5821/booking/select-seat/1
MÀN HÌNH
Ghế đang chọn: 2 ghế
TIẾP TỤC THANH TOÁN


localhost:5821/booking-v2/summary?seat=B6&loungeId=1&food=1
XÁC NHẬN THANH TOÁN
Mã đơn hàng: #33
Trạng thái: PENDING
Tổng thanh toán: 75,000 VNĐ
09:55
Quét mã VietQR
Mở App Ngân hàng bất kỳ để quét mã
VIETQR
napas 247
MB
Ngân hàng: MBBank (MB)
Chủ TK: LƯU DƯƠNG BẢO BẢO

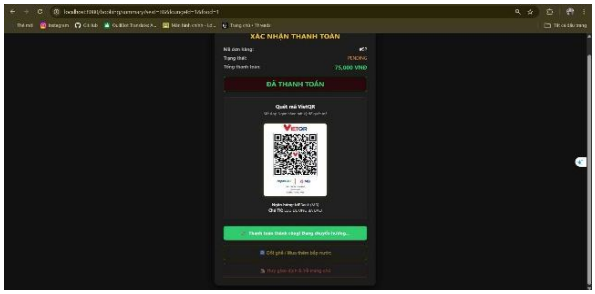
Payment Processing Flow

No.	Flow Step	Test Scenario
1	Open Booking Summary	User proceeds to checkout
2	Calculate Total Price	System calculates ticket and food cost
3	Payment Request	User confirms payment
4	Payment Confirmation	System receives payment webhook
5	Ticket Generation	System generates digital ticket

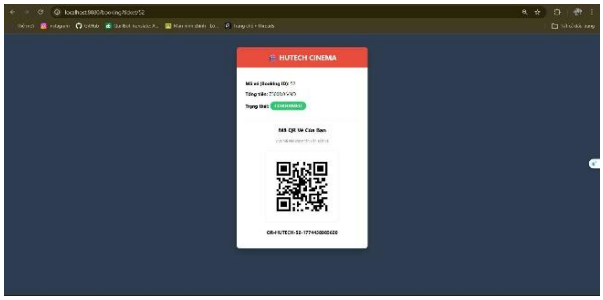
Payment Flow

Input	Expected Result	Status
BookingID: 52	System displays booking summary	Pass
Ticket: 75000, Food: 0	Total price = 75000	Pass
BookingID: 52	System sends payment request to payment gateway	Pass
Content: “HUTECH 52”	Booking status updated to CONFIRMED	Pass
BookingID: 52	Digital ticket with QR Code is generated	Pass

Image



booking_id	user_id	showtime_id	total_price	status	qr_code_token	created_at	expires_at
50	1	1	75000.00	CANCELLED	NULL	2026-03-25 23:26:28	2026-03-25 16:31:28
51	4	1	75000.00	CANCELLED	NULL	2026-03-26 00:15:16	2026-03-25 17:20:17
52	4	1	75000.00	CONFIRMED	QR-HUTECH-52-1774458985620	2026-03-26 00:16:04	2026-03-25 17:21:05



Seat Release Flow

No.	Flow Step	Test Scenario	Input
1	Booking Timeout	Booking exceeds holding time	BookingID: 32
2	Cancel Booking	System cancels expired booking	BookingID: 32
3	Release Seats	System releases locked seats	SeatIDs: [E45]
4	Update Seat Status	Database updates seat availability	SeatIDs: [E45]

Expected Result	Status
System detects expired booking	Pass
Booking status updated to CANCELLED	Pass
Seats become available again	Pass
is_available = 1	Pass

Image

booking_id	user_id	showtime_id	total_price	status	qr_code_token	created_at	expires_at
25	1	1	75000	CANCELLED	NULL	2026-03-26 23:38:23	2026-03-26 23:48:24
26	1	1	75000	CANCELLED	NULL	2026-03-26 23:38:32	2026-03-26 23:48:33
27	1	1	75000	CANCELLED	NULL	2026-03-26 23:51:48	2026-03-27 00:01:49
28	1	1	75000	CANCELLED	NULL	2026-03-26 23:51:55	2026-03-27 00:01:55
29	1	1	75000	CANCELLED	NULL	2026-03-26 23:54:00	2026-03-27 00:04:01
30	1	1	75000	CANCELLED	NULL	2026-03-27 13:06:40	2026-03-27 13:16:40
31	1	1	75000	CANCELLED	NULL	2026-03-27 14:54:31	2026-03-27 15:04:31
32	1	1	75000	CANCELLED	NULL	2026-03-27 15:04:07	2026-03-27 15:14:07

seat_id	lounge_id	seat_number	row_char	is_available
44	1	44	E	1
45	1	45	E	1

User Authentication Flow (Spring Security)

No.	Flow Step	Test Scenario
1	Access Protected Page	User tries to access booking page
2	Login Request	User submits login form
3	Authentication Success	Valid credentials
4	Role Verification	Admin accesses admin page
5	Unauthorized Access	Normal user accesses admin page

Input	Expected Result	Status
URL: /booking/select-seat	Redirect to login page	Pass
Email + Password	System authenticates user	Pass
User account	Redirect to movie page	Pass
Admin account	Access granted to admin dashboard	Pass
User account	Access Denied page displayed	Pass

Test Result Summary (Unit Test & Integration Test)

Test Type	Module / Function	Total Test Cases
Unit Testing	Showtime & Lounge Selection	3
Unit Testing	Seat Selection	5
Unit Testing	Booking Summary & Payment	5
Unit Testing	Spring Security Authentication	5
Integration Testing	Movie Booking Flow	6
Integration Testing	Seat Reservation Flow	5
Integration Testing	Payment Processing Flow	5
Integration Testing	Seat Release Flow	4
Integration Testing	Security Authentication Flow	5

Overall Test Result

	Total Test Cases	Passed
Unit Testing	18	18
Integration Testing	25	25
Total	43	43

Passed	Failed	Pass Rate
3	0	100%
5	0	100%
5	0	100%
5	0	100%
6	0	100%
5	0	100%
5	0	100%
4	0	100%
5	0	100%

Failed	Pass Rate
0	100%
0	100%
0	100%

Notes

All functions executed correctly
Seat retrieval and seat locking work properly
Payment processing and booking confirmation successful
Authentication and authorization working as expected
End-to-end booking flow executed successfully
Seat selection and temporary locking functioning correctly
Payment confirmation and ticket generation successful
Expired bookings correctly release seats
Unauthorized access is properly restricted